MEASURES OF CENTRAL TENDENCY

DESCRIPTIVE STATISTICS

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

LABORATORY

50%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Mean

Median

Mode

MEAN

POPULATION AND SAMPLE

Population

Collection of **all** items of interest, $\boldsymbol{N}$

Sample

A **subset** of population, $\boldsymbol{n}$

POPULATION AND SAMPLE

Population

Collection of **all** items of interest, $\boldsymbol{N}$

Sample

A **subset** of population, $\boldsymbol{n}$

parameters – data obtained from population

statistics – data obtained from a sample

MEAN

Population	Sample
μ	$\bar{x}$

The arithmetic **center**

a.k.a a simple average

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{n}$$

Determine the **mean** of the given pizza prices.

Pizza Prices

Position	New York City	Los Angeles
1	\$ 1	\$ 1
2	\$ 2	\$ 2
3	\$ 3	\$ 3
4	\$ 3	\$ 4
5	\$ 5	\$ 5
6	\$ 6	\$ 6
7	\$ 7	\$ 7
8	\$ 8	\$ 8
9	\$ 9	\$ 9
10	\$ 11	\$ 10
11	\$ 66	

New York City

$$\bar{x} = \frac{\mathbf{1+2+3+3+5+6+7+8+9+11+66}}{11}$$

$$\boxed{\bar{x} = \$\,\mathbf{11}}$$

ans

Los Angeles

$$\bar{x} = \frac{\mathbf{1+2+3+4+5+6+7+8+9+10}}{10}$$

$$\boxed{\bar{x} = \$\,\mathbf{5.5}}$$

ans

MEDIAN

MEDIAN

The **midpoint** of the ordered data set

Median is at position

$$\frac{n+1}{2}$$

Determine the **median** of the given pizza prices.

Pizza Prices

Position	New York City	Los Angeles
1	$ 1	$ 1
2	$ 2	$ 2
3	$ 3	$ 3
4	$ 3	$ 4
5	$ 5	$ 5
6	$ 6	$ 6
7	$ 7	$ 7
8	$ 8	$ 8
9	$ 9	$ 9
10	$ 11	$ 10
11	$ 66	

New York City

Median is at position

$$\frac{n+1}{2}$$

$$\frac{11+1}{2} = 6^{th}\ position$$

$$\boxed{\text{Median} = 6}$$

ans

Determine the **median** of the given pizza prices.

Pizza Prices

Position	New York City	Los Angeles
1	$ 1	$ 1
2	$ 2	$ 2
3	$ 3	$ 3
4	$ 3	$ 4
5	$ 5	$ 5
6	$ 6	$ 6
7	$ 7	$ 7
8	$ 8	$ 8
9	$ 9	$ 9
10	$ 11	$ 10
11	$ 66	

Los Angeles

Median is at position

$$\frac{n+1}{2}$$

$$\frac{10+1}{2} = 5.5^{th}\ position$$

$$\text{Median} = \frac{5+6}{2}$$

$$\boxed{\text{Median} = 5.5}$$

ans

MODE

MODE

The mode is the value that occurs **most often**

A dataset can have 0 mode, 1 mode or multiple modes

1
1
1
2
2

Count = 3

Determine the **mode** of the given pizza prices.

Pizza Prices

Position	New York City	Los Angeles
1	$ 1	$ 1
2	$ 2	$ 2
3	$ 3	$ 3
4	$ 3	$ 4
5	$ 5	$ 5
6	$ 6	$ 6
7	$ 7	$ 7
8	$ 8	$ 8
9	$ 9	$ 9
10	$ 11	$ 10
11	$ 66	

New York City

Mode = 3

ans

Los Angeles

NO mode

ans

LABORATORY